

Nikhil Dixit

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Education

Vellore Institute of Technology	2022 – 2026
B. Tech in Computer Science & Engineering	CGPA: 8.49/10
Bhartiya Vidya Bhavan's Mehta Vidyalaya, New Delhi (CBSE)	2008 – 2021
Intermediate XII (Senior-secondary)	85.1%
High School X (Secondary)	93.3%

Technical Skills

- **Programming & Data Handling:** Python, Pandas, NumPy
- **Machine Learning & Visualization:** Scikit-learn, Matplotlib, Seaborn
- **Generative AI & NLP:** LangChain, Hugging Face, Prompt Engineering, Embeddings
- **Databases & Retrieval Systems:** FAISS, SQL
- **Document Processing & Developer Tools:** PyMuPDF, Git

Experience

Digital Intern	Jun 2024 – Sep 2024
KPMG India (Digital Lighthouse)	Python, Pandas, LangChain, SQL, PyMuPDF
• Developed Python scripts using Pandas to automate extraction and cleaning of unstructured PDF reports • Explored Generative AI frameworks (LangChain) to evaluate feasibility for internal summarization • Executed SQL queries on to extract specific client records, ensuring data accuracy for weekly reports. • Documented code workflows and created user manuals to enable non-technical teams to use automation tools	

Projects

DocuSearch: AI-Powered Q&A System	May 2025 – Jul 2025
RAG Chatbot with Flan-T5	Python, LangChain, Hugging Face, FAISS
• Designed and implemented an end-to-end Retrieval-Augmented Generation (RAG) system using Flan-T5-Large, achieving over 90% factual accuracy • Built a scalable ingestion pipeline to process PDF documents and generate optimized text chunks for embeddings • Integrated a FAISS vector database to enable low-latency semantic search, reducing retrieval time to under 500 ms while ensuring grounded responses	

Heart Attack Risk Prediction System	Jan 2025 – Mar 2025
Machine Learning Project	Python, Scikit-learn, Pandas, NumPy, Matplotlib
• Built a supervised classification model to predict heart attack risk using clinical and lifestyle features • Performed data preprocessing and feature engineering, including normalization and class imbalance handling • Evaluated Logistic Regression and Random Forest models, achieving 88% accuracy on test data	

EstatePredict: Real Estate Market Prediction Tool	Oct 2024 – Nov 2024
Machine Learning Project	Python, Scikit-learn, Pandas, NumPy, Seaborn
• Implemented a comprehensive data preprocessing workflow for 5,000+ property listings, including encoding and feature scaling • Trained and evaluated Linear Regression ($R^2=0.80$), Decision Tree ($R^2=0.86$), and Random Forest ($R^2=0.91$) models • Selected the Random Forest model as best-performing based on predictive accuracy and error metrics	

Achievements and Certifications

- Competed for New Delhi District U17 Football Team
- Effective Communication in the Workplace
- IBM Generative AI Engineering Professional